JAGADISH GOSALA

Senior Full-Stack Engineer

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SUMMARY

Full-stack engineer who leads a 4–6 engineer team and owns two production platforms at Provakil. Built a self-serve analytics platform — UI-composed filters and groupings compiled into MongoDB aggregation pipelines, executed asynchronously — and re-architected it from 2.5M to 15M records/day (~6x). Also built an organisation-wide LLM gateway with usage metering, PII masking, and chargeback, and shipped a multi-tenant SaaS product solo. Currently building a RAG legal research assistant with hybrid retrieval, paragraph-level citations and a measured evaluation harness. 5 years across backend, data, and product engineering.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, SQL
Backend: Node.js, Express.js, Sails.js, FastAPI
Frontend: Vue.js, Next.js
Data: MongoDB (aggregation pipelines, index design), PostgreSQL, MySQL, Redis, Elasticsearch, Qdrant
AI & Retrieval: RAG, Hybrid search (dense + BM25, RRF), BGE-M3 embeddings, Cross-encoder reranking, Ollama, Retrieval evaluation (MRR, hit@k), PyMuPDF (PDF extraction), Presidio (PII masking)
AI Coding Agents & LLM APIs: Claude Code, Claude (agentic workflows), OpenAI Codex, GitHub Copilot, Gemini API, DeepSeek API
Infrastructure: Docker, Docker Compose, RabbitMQ, AWS (S3, EC2), MinIO, Supabase, Vercel, WebSockets, Alembic, Cypress, pytest, CI/CD

PROFESSIONAL EXPERIENCE

PROVAKIL — Software Developer, Team Lead **May 2023 – Present**
Enterprise legal-tech SaaS for banks and insurers

Platform Ownership & Engineering Leadership

- Lead a 4–6 engineer team owning the analytics platform end to end — architecture decisions, design reviews, code review, and on-call incident response. Interview candidates and mentor juniors through onboarding.
- Re-architected the pipeline from 2.5M to 15M records/day (~6x): pushed filter predicates into the join stage instead of filtering post-join, and added compound indexes derived from cross-collection access patterns after confirming write amplification on source collections was negligible.
- Replaced full-pipeline recomputation with incrementally-refreshed materialized views for the most-requested report types — a source-document update enqueues a single-record refresh, trading configurable dynamic fields for near-real-time freshness. Unblocked a client expansion of 200+ enterprise users.
- Partitioned processing into scheduled-refresh, email-delivery, and long-running (>1 hr) workload classes on independently scaled consumers, isolating slow aggregations so they no longer block routine refreshes.
- Own and extended a metadata-driven entity platform (AnyCabinets) where new object types — contracts, obligations, patents — are defined in stored configuration rather than code, comparable to Salesforce custom objects. Built a bank-specific contract-review and checklist workflow on top of it with AI-assisted extraction.
- Classified every inbound support ticket over six months to build a root-cause picture of the module, then eliminated the dominant failure classes in query logic and data sync — cutting ticket inflow ~70% (~15 ops-hours/week).
- Built the platform's Vue.js interface — filter builders, dynamic charts, and data tables driven by the same configuration model as the backend.

Backend Services & Infrastructure

- Built an organisation-wide LLM gateway (FastAPI) fronting every model call — wallet/ledger credit accounting, per-team and per-developer limits, PII masking before requests leave the network, provider routing, and rate-card chargeback. Ships as its own container for cloud and on-prem.
- Built an e-signature and statutory stamp-duty platform from scratch — provider integrations, document state machine, a dead-letter queue for failed requests, and environment-aware routing to S3 or MinIO.
- Built the customer-onboarding bulk-import pipeline — Python consumers coordinating via atomic Redis counters to emit throttled WebSocket progress events across parallel importers.

- Instrumented monitoring across the reporting stack — crashed-job detection, cron and generation-failure alerting, and Cypress regression coverage for the document pipeline.
- Built a production-to-on-prem migration tool reconciling cross-environment schema differences, enabling 3 enterprise on-prem deployments.

Applied LLM Systems

- Built a structured-output clause-rewrite tool that reads a contract together with its email correspondence and proposes redlines for review.
- Built multi-stage email automation — an LLM links inbound threads to the correct case record and chains modules so a conversation advances a matter from pre-litigation to litigation without manual data entry.
- Built a multi-language translation pipeline for 8+ languages including right-to-left scripts (Arabic, Hebrew, Urdu) — layout-direction switching, font fallbacks, and formatting preservation; cut per-document cost ~85% via prompt caching and batching.
- Built Python summarisation and metadata-extraction pipelines over documents running 50+ pages — removed 2–3 hours of manual reading per case and cut ops-team tagging by 50%+ (previously 40+ man-hours/day). Took an extraction proof-of-concept for daily court docket PDFs — inconsistent layouts, no fixed schema — into a production ingestion pipeline.

RJEYS INFINITY SOLUTIONS — Assistant Solutions Engineer **April 2021 – May 2023**

Software services firm

- Built backend APIs and production debugging workflows on a Node.js/Express service serving 10K+ daily active users, plus a client data-migration tool handling schema mapping, validation, and edge cases for 500K+ legacy records.
- Served as technical analyst for Gainsight; rewrote heavy automated batch jobs against MySQL with optimised queries and indexing — cut runtimes by 60%.

PROJECTS

LEGAL CASE RESEARCH ASSISTANT (RAG) — sole engineer **Python · FastAPI · Qdrant · Next.js**

- Building a RAG research assistant over Indian Supreme Court judgments that cites the exact judgment and paragraph. Wrote the PDF ingestion pipeline (PyMuPDF): geometry-based removal of reporter margin letters, headnote/judgment splitting, and paragraph-numbering rules tolerant of OCR errors — raised numbered-paragraph detection from 66% to 96% of judgments; 500 judgments indexed as 19,914 chunks.
- Built hybrid retrieval in FastAPI over Qdrant — BGE-M3 dense embeddings plus BM25 sparse vectors fused with reciprocal rank fusion, and a two-stage citation mode that picks judgments on editor headnotes, then selects paragraphs within them. Every model sits behind a provider interface, so local (Ollama) and hosted inference switch by configuration.
- Wrote a 50-question golden set and evaluation harness (document hit@k, MRR, paragraph hit, latency) and tuned against it: MRR@10 0.953 with the correct judgment in the top 3 for every question, paragraph hit@3 up from 0.46 to 0.60, and p50 query latency cut from 375 ms to 37 ms. Dropped the cross-encoder reranker after the eval showed no ranking gain at 10 s+ CPU latency.
- Sized it for ₹0/month free-tier hosting: float16 vectors on disk with an int8 in-RAM copy project the ~10K-judgment corpus to ~2.9 GB of the 4 GB vector-DB free tier.

CAMPUS PLACEMENT PLATFORM — sole engineer **hyrd.co.in**

- Built a multi-tenant SaaS platform for university recruiting — four role types with hierarchical admin sub-roles, access control enforced through Postgres row-level security policies rather than in application code, and verified tenant isolation.
- Built an AI resume-tailoring workspace that scores a resume against a specific job description and company profile, returns per-section findings with reasoning, and generates a rewrite presented as a live diff — all under explicit anti-fabrication constraints and per-user rate limits.
- Shipped bulk CSV/Excel import with column mapping and validation, magic-link invites, audit logging, transactional email, web-push/PWA notifications, and PDF/Excel reporting. Stack: Next.js, TypeScript, PostgreSQL (Supabase), Vercel.

EDUCATION

B.Tech, Electronics and Communications Engineering — RGUKT **2017 – 2021**